

**Export Feasibility of Wooden Door Manufacturing: A Comparative Location Analysis of
India, Canada, and Nepal for European and North American Markets**

Jenny Barreto, Tiana Bender-Santiago, Ezeoma Igwe, Mia Moore, Gurdev Singh, Joseph

Muganza

BUS511 – A – FiNEcnmc Strtgy

California Baptist University

Dr. Alex Sherm

March 3, 2026

Table of Contents

<i>Executive Summary (Lewis)</i>	6
<i>Introduction (Mia and Tiana)</i>	7
Background and Rationale (Mia)	7
Project Concept (Mia)	10
Scope and Limitations (Tiana)	11
<i>Market Feasibility</i>	12
Global Furniture Market Overview (Gurdev)	12
Country Selection Logic	14
India (Gurdev)	14
Canada (Joseph)	14
Nepal (Jenny)	15
Demand and Customer Segments (Gurdev)	15
Comparative Market Position by Country	16
India (Gurdev)	16
Canada (Joseph)	16
Nepal (Jenny)	17
Competitive Landscape (Gurdev)	18
Logistics and Trade Routes	20
Canada (Joseph)	20
India (Gurdev)	21

Nepal (Jenny)	22
Regulatory and Standards Considerations.....	23
India (Gurdev).....	23
Canada (Joseph)	23
Nepal (Jenny)	24
<i>Technical Feasibility.....</i>	<i>24</i>
CANADA (Joseph).....	24
Production Process Description	24
Production Capacity Logic	25
Technology and Equipment	26
Location, Infrastructure, Labor, and Skills	27
NEPAL (Joseph).....	27
Production Process Description	27
Production Capacity Logic	28
Technology and Equipment	29
Location, Infrastructure, Labor, and Skills	29
INDIA (Gurdev)	30
Production Process Description	30
Technology and Equipment	30
Location, Infrastructure, Labor, and Skills	31
<i>Financial Feasibility and Analysis.....</i>	<i>31</i>
NEPAL (Jenny)	31

Investment Requirements.....	31
CANADA (Jenny)	39
Investment Requirements.....	39
INDIA (Gurdev)	44
Investment Requirements.....	44
Machinery Assumptions	45
Production Schedule	45
Bottleneck-Based Capacity Calculation	45
<i>Economic and Sensitivity Analysis.....</i>	52
CANADA (Joseph and Mia)	52
Economic Impact (Mia)	52
Sensitivity Analysis (Joseph).....	53
Scenario Discussion (Joseph)	55
Standardized Sensitivity Variables	55
NEPAL (Jenny)	56
<i>Risk Assessment and Mitigation (Mia & Gurdev)</i>	57
Country-Specific Risk Analysis	58
India (Mia and Gurdev)	58
Nepal (Mia).....	59
Canada (Mia)	62
<i>Strategic and Ethical Considerations (Tiana)</i>	64

<i>Conclusion and Recommendations (Lewis)</i>	66
Summary of Findings	66
India	66
Nepal	66
Canada.....	67
Recommendations	68
Comparative Summary Table	68
Recommendation strategy (Joseph)	69
Conditions and Next Steps	70
<i>References</i>	72

Executive Summary (Lewis)

This feasibility study evaluates whether an export-oriented wooden door manufacturing operation can competitively serve European and North American markets, and which production location offers the most attractive balance of cost, risk, and sustainability. This feasibility study recommends India as the optimal location for establishing an export-oriented wooden door manufacturing operation serving European and North American markets. While Canada and Nepal are financially and technically viable, India provides the strongest overall balance of return, scalability, and risk management. The economic factors considered in evaluating the chosen countries include each country's production capacity, political risk, exchange rates, raw materials resources, cost competitiveness, ability to meet international quality, environmental and labor standards. The purpose of this study is to identify the most viable production and in which of the three countries may be most attractive based on supplying competitive, sustainable and high-quality furniture products to global markets.

This project will evaluate the financial feasibility using indicators such as Net Present Value (NPV), Internal Rate of Return (IRR), Return on Investment (ROI), and Payback Period. Although Canada generates the highest Net Present Value, India outperforms on capital efficiency and speed of return. The IRR of 24% and short payback period of 2.3 years indicate faster recovery of invested capital and stronger annualized profitability. From an investment perspective, IRR and payback are critical measures of risk-adjusted return and liquidity. India therefore dominates superior return intensity short payback period of 2.3 years indicate faster recovery of invested capital and stronger annualized profitability.

Technically, India benefits from skilled labor, semi-automated production capacity, competitive operating costs, and established export infrastructure. Canada offers advance

automation, premium sustainability positioning, and reduced trade barriers under agreements such as the United States-Mexico-Canada Agreement and the Comprehensive Economic and Trade Agreement. Nepal demonstrates moderate financial returns supported by low labor costs but faces structural constraints, including high logistical costs, policy uncertainty, and limited automation.

The trade-offs are clear: Canada provides regulatory stability and sustainability credibility but at higher production costs; Nepal offers cost advantages but with elevated logistics and supply chain risk. India carries exposure to exchange rate and raw material volatility, yet sensitivity analysis confirms resilience under adverse scenarios.

Finally, the study illustrates that export-oriented furniture production is feasible for Europe and North America with adequate strategies implemented. India delivers the highest IRR at 24% with a payback of approximately 2.3 years, outperforming Canada's 19% IRR and Nepal's 21% IRR, while maintaining manageable supply chain risk.

Introduction (Mia and Tiana)

Background and Rationale (Mia)

The global construction market has experienced sustained growth over the years, driven by key factors such as urbanization, population growth, and increased housing development. These trends have also been observed across numerous developed and emerging economies, including Canada, India, Nepal, and the United States of America (U.S.). These markets represent a diverse cross-section of global demand. As construction activity continues to grow, it is vital to continue studying the need for residential and commercial building materials, including products based on natural material, wood. While this feasibility study examines export-oriented

furniture manufacturing broadly, it focuses specifically on wooden doors as a representative product category, given their strong link to construction demand and regulatory standards.

Throughout the construction and building materials sector, wooden doors have been a critical part of the product category. This demand stems from their function, aesthetic value, and overall role in residential and commercial buildings. Although many consumers have different preferences for door structure, material, color, and style, wooden doors have been widely implemented globally due to institutional regulations and renovation cycles. Depending on the region, wooden doors must follow standardized dimensions and comply with fire, safety, and insulation regulations. These products are installed in both government buildings and residential spaces and serve the needs of new construction and renovation projects. Because of these standardized and regulatory characteristics, construction-linked products typically enable more reliable demand forecasting when using macroeconomic indicators (Lee, 1994).

Demand for wooden doors has recently been strong in North American and European markets (Melati et al., 2013). Recent construction activity has further supported the concept that there is a demand for a wooden export-oriented business model. According to a University of Oregon study using econometric analysis, wooden doors are historically sensitive to changes in residential construction prices, with an annual elasticity of 3.54% (Lee, 1994). While market dynamics have evolved due to socioeconomic, construction, and economic factors (Zhou, 2023), this analysis suggests that there may be market feasibility for exporting wooden doors.

According to the U.S. Census Bureau, recent construction spending data indicate that the total United States construction spending has consistently remained over \$2.1 trillion annually, with the most recent data as of February 2026 currently valued at \$2.17 trillion (U.S. Census Bureau, 2026). One particular opportunity lies in the \$521 billion public sector and the \$742

billion private nonresidential segment, representing schools, hospitals, government offices, and institutional procurement. In addition, residential construction spending has increased steadily each month. According to a formal analysis done by the National Association of Home Builders (NAHB) utilizing data from the U.S. Census Bureau's Survey of Construction, the average building size begins between 1,600 and 3,000 sq. ft. Assuming that each building has an approximate minimum of 12 interior and exterior doors, with at least 940,000 single-family building starts in 2023 as a baseline, the doors can target at least .13% of the annual new construction demand (NAHB, 2024; U.S. Census Bureau, 2026)

This data shows that residential construction and renovation have been an important driver of demand for building materials (U.S. Census Bureau, January 2026). In addition, construction activity has remained elevated despite market fluctuations, thereby reinforcing the idea that doors will retain long-term relevance in the market. The combination of demand and supply chain factors has created an opportunity for a potential door manufacturing and export business.

Wooden doors were chosen as the export topic of the feasibility study due to their direct correlation with construction activity, regulatory requirements, and alignment with sustainability through FSC (architectural-grade) certification. The product category also maintains a strong market presence across B2B and B2G channels (Zhou, 2023). Wooden doors are structural necessities governed by building codes and closely align with macroeconomic construction indicators. By combining these indicators with financial forecasting, regulatory compliance assessment, and export feasibility modeling, a study can and should be pursued.

This feasibility research paper integrates various disciplines to assess whether wooden door manufacturing and exporting can be efficiently and successfully positioned for international

markets. Some examples include business analytics, financial and economic strategy and modeling, and scenario-based decision-making. Essentially, this feasibility study will also serve as a tool for linking qualitative and quantitative decision-making, in combination with cost and sensitivity analyses.

The study will also incorporate elements of decision-making through an ethical and biblical lens by taking into account socioeconomic barriers to entry, sourcing, and data transparency. Initial group findings indicate that the project is feasible. However, final recommendations will be made at the conclusion of the study, after capturing financial indicators such as NPV, IRR, and payback period, as well as operating costs and projections for each country.

Project Concept (Mia)

The purpose of this feasibility study is to determine whether an export-oriented wooden door manufacturing project creates sufficient economic value to justify capital investment, and to identify which of the three proposed production locations, Canada, India, or Nepal, provides the strongest risk-adjusted financial and strategic outcome.

Because furniture manufacturing is capital-intensive, logistics-sensitive, and exposed to international regulatory standards, location selection directly shapes revenue potential, cost structure, operating volatility, and long-term competitiveness (Ferdows, 1997). Differences in labor costs, timber access, certification systems, infrastructure reliability, and trade agreements influence not only operating margins but also the stability of projected cash flows (OECD, 2023; ITC, 2022).

Accordingly, this study integrates qualitative market assessment with standardized quantitative financial modeling under consistent assumptions of production capacity (15,000

units annually), project life (10 years), and discount rate (10%). Financial feasibility is evaluated using Net Present Value (NPV), Internal Rate of Return (IRR), and Payback Period to compare capital efficiency, value creation, and risk exposure across country scenarios.

While positive NPV indicates economic value creation, IRR and payback period provide insight into capital recovery speed and sensitivity to operational risk. Therefore, the final recommendation is not based solely on absolute NPV, but on a balanced interpretation of capital exposure, return velocity, operating volatility, and long-term scalability.

By structuring the analysis around comparable assumptions and consistent financial metrics, this study supports a disciplined capital allocation decision rather than a purely descriptive location comparison.

Scope and Limitations (Tiana)

Despite its structured methodology, this feasibility study represents a strategic screening analysis rather than a final investment authorization. Several limitations should be acknowledged prior to capital commitment.

Standardized production assumptions. All country models are constructed using identical production capacity (15,000 units annually) and project life (10 years) to enable direct financial comparability. However, pricing assumptions reflect different market positioning strategies, premium positioning in Canada, mid-range positioning in India, and value-to-mid positioning in Nepal. Consequently, cross-country financial results reflect both cost structure differences and segment-specific pricing strategies.

Macroeconomic stability assumptions. Exchange rates, energy costs, interest rates, and trade policies are assumed to remain relatively stable over the modeling horizon. Significant

currency volatility, geopolitical disruption, or regulatory shifts could materially alter projected cash flows and effective discount rates.

Reliance on secondary data. The analysis draws upon published data from international institutions and trade sources. Variations in reporting standards, particularly in emerging markets, may limit precision in cross-country comparisons (World Bank, 2023). To mitigate this limitation, multiple sources were triangulated where possible.

Regulatory simplification. While major environmental, labor, and certification standards are incorporated, the study does not model firm-specific compliance costs or potential future policy changes. Detailed legal and regulatory due diligence would be required prior to implementation.

Sensitivity modeling constraints. Although sensitivity testing is conducted, not all variables receive identical stress testing across scenarios. Additional modeling of freight volatility, raw material shocks, and exchange-rate fluctuations would further strengthen risk assessment.

The findings should be interpreted as a structured feasibility framework designed to evaluate relative attractiveness under comparable financial assumptions. Final investment decisions would require detailed engineering studies, contractual supplier commitments, and enhanced macroeconomic risk modeling.

Market Feasibility

Global Furniture Market Overview (Gurdev)

The global furniture market has experienced steady expansion in recent years, driven by rapid urbanization, population growth, rising disposable incomes, and evolving consumer lifestyles (Statista, 2024). Developed regions such as Europe and North America continue to

represent the largest and most stable furniture markets worldwide, accounting for a substantial share of global consumption. Demand across these regions spans residential, office, and commercial segments, including wooden doors for housing developments, corporate offices, hospitality establishments, and public infrastructure projects.

Similarly, the international wooden doors market has recorded consistent growth, supported by expanding housing construction, renovation activities, and commercial real estate development. Rising income levels and improved living standards in both developed and emerging economies have further strengthened demand. Europe and North America remain key consumption hubs, characterized by strong import reliance to meet diverse design preferences, cost competitiveness, and supply chain flexibility.

Changing consumer behavior has significantly influenced product demand patterns. The expansion of remote and hybrid work models has increased investment in home improvement and home office infrastructure, including interior wooden doors designed for privacy, sound insulation, and aesthetic integration. At the same time, modern urban lifestyles have encouraged preference for minimalist, functional, and space-efficient designs. Modular and ready-to-assemble wooden doors have gained popularity due to their ease of transportation, reduced logistics costs, and simplified installation processes—factors that align well with contemporary distribution channels and e-commerce growth.

Sustainability considerations now play a central role in purchasing decisions, particularly in developed markets. Consumers and regulators increasingly expect manufacturers to use certified sustainable timber, low-emission finishes, and environmentally responsible production methods. Compliance with environmental, labor, and traceability standards has therefore become a prerequisite for market entry. International policy frameworks and regulatory guidance from

organizations such as the Organisation for Economic Co-operation and Development emphasize responsible sourcing, supply chain transparency, and sustainable production practices (OECD, 2023). Consequently, exporting countries and manufacturers seeking to access European and North American markets must align their operations with stringent environmental certifications and social compliance standards to remain competitive.

Overall, the wooden doors segment reflects broader structural shifts within the global furniture industry—where growth is increasingly shaped by sustainability expectations, design innovation, urban housing expansion, and evolving work patterns.

Country Selection Logic

India (Gurdev)

India, Canada, and Nepal were selected for this study because they represent different economic contexts, production capabilities, and export strategies within the global wooden door industry. India is one of the world's fastest-growing manufacturing economies and has a large labor force with relatively low wage levels. The country has increasing experience in exporting manufactured goods and is actively promoting industrial growth and export-oriented production (World Bank, 2023). Its geographic location and access to multiple major seaports make it suitable for serving distant international markets.

Canada (Joseph)

Canada was selected for the feasibility analysis given its status as a world leader in forest industry and water-based furniture wood inputs, with well-managed and abundant forest resources combined with strong manufacturing capabilities. Canada has almost 39% of the land covered by forest in the world, and approximately 9 % of the world's total forest cover, an asset that represents a powerful, solid wood manufacturing industry essential for furniture production. Canada Among World's Largest Producers and Exporters of Wood Products (WC0073) Canada's

broad base of forest resources have become a major economic driver, accounting for a significant portion of national GDP and employment through forestry and value-add wood manufacturing.

Canadian forest products such as timber and wood- used furniture related components are widely exported due to their well-established management system and international transactions, which means that Canada is a potentially important country when considering the feasibility of export-oriented furniture production. While Canada has a broad furniture manufacturing base, this feasibility analysis focuses especially on wooden doors as a high-value, construction-linked product category supported by Canada's certified timber supply and advanced wood-processing capability.

Nepal (Jenny)

Nepal is considered an emerging-economy opportunity within the wood processing and manufacturing sector due to low labor costs, woodworking traditions, and growing participation in global value chains. This analysis reviews existing manufacturing capabilities, resources, supply chain, industrial infrastructure, technology, human resources, and more to determine whether Nepal is an ideal country of production for global markets.

Demand and Customer Segments (Gurdev)

The primary target markets for wooden doors exports from India, Canada, and Nepal are Europe and North America, which together represent a large and relatively stable source of demand (Statista, 2024). These markets import significant volumes of wooden doors each year to meet consumer demand that cannot be fully satisfied by domestic production. The main customer segments include wholesale distributors that supply wooden doors to regional and national retailers, large retail chains that sell wooden doors directly to consumers, and online platforms that focus on convenience and design-oriented products. In addition, some producers may target specialized design brands that emphasize sustainability, craftsmanship, or customization. Across

these segments, customers generally value consistent product quality, competitive pricing, reliable delivery schedules, and the ability of suppliers to meet international safety and environmental standards. Sustainability and ethical production practices have become particularly important, as many buyers are under pressure from consumers and regulators to ensure responsible sourcing. Export-oriented wooden door producers must therefore align their production processes and supply chains with these expectations in order to succeed.

Comparative Market Position by Country

India (Gurdev)

India's market position is supported by its large manufacturing base, availability of skilled and semi-skilled labor, and relatively low production costs. The country has several major seaports that connect it efficiently to Europe and North America, making large-scale exports feasible. India also benefits from growing experience in meeting international buyer requirements. However, challenges remain, including infrastructure constraints, regulatory complexity, and variations in quality standards across producers. Table 1 illustrates the comparison between the countries.

Canada (Joseph)

Canada is well positioned in the global furniture market as it competes in high-value segments of quality, sustainability, and regulatory compliance. The countries have relatively high shares of certified forests, which help ensure that purchased wood enters a system of responsible sourcing and environmentally sustainable management (Natural Resources Canada, 2023). These sustainability credentials are well-matched with increasing demand in developed markets for environmentally certified furniture products. Moreover, there is an existing export presence as Canada's furniture exports amounted to around \$3.2B USD (2023), which are mostly directed toward the United States market, demonstrating that Canadian firms can access international

supply chains and maintain regular buyer relationships (OEC 2024). With the elimination of most industrial tariffs and improved market access to Europe, Canada's CETA-supported competitive advantage is enhanced, further promoting this country as a trustworthy provider of quality and sustainability-oriented furnishings (Global Affairs Canada, 2023). These factors position Canada as the most suitable production location for premium, regulation-sensitive wooden doors, where quality assurance and sustainability outweigh cost considerations.

Nepal (Jenny)

Both local artisans and larger industrial firms are established in Nepal but face operational hardships due to government regulations. According to the national industrial survey, there are approximately 1,679 wood product manufacturing businesses in the country, including door and related wood products. Approximately 40% of the Rs 50 billion in Nepal's furniture business consists of raw materials from the domestic market; the other 60% is outsourced. Due to regulatory constraints on timber harvesting, Nepal's wood manufacturing is dependent on imported wood. Despite the country being covered by approximately 42% forest, the preferred type required for exterior doors is hardwood, and this accounts for only approximately 59% of the forest resources. The forest conservation policy prohibits cutting trees for profit, leading to smuggled wood for construction (New Business Age, 2026).

Business leaders in Nepal are urging the government to improve the balance between forest protection and industrial development. Nepal's furniture market is expanding rapidly due to the construction and industry boom, driven by growth in real estate, hospitality, housing, and rebuilding of structures damaged by earthquakes. The government has been pressured to require public offices to prioritize the use of at least 50% locally sourced products when available to promote "Made-in-Nepal" goods, reduce the trade deficit, retain foreign currency, and provide a better market space for domestic furniture manufacturers. (New Business Age, 2026).

Table 1:*Comparative Market Position for Wooden doors Exports*

Factor	India	Canada	Nepal
Labor cost competitiveness	High	Low	High
Access to seaports	High	High	Low
Export experience	High	High	Low–Medium
Business environment	Improving	Stable	Improving

Note: For the Labor cost competitiveness, High: Average manufacturing wage >40% lower than OECD industrial benchmark, Medium: 15–40% lower than OECD benchmark and Low: <15% lower than OECD benchmark. Access to seaports, High: Major international port access within 300 km; average export transit time <25 days; reliable container capacity, Medium: Port access 300–800 km; transit time 25–40 days; moderate congestion risk, and Low: Limited or no direct seaport access; reliance on third-country ports; transit time >40 days. For Export experience, High: >10 years of sustained wood product exports; diversified export destinations; established compliance systems, Medium: 5–10 years export experience; moderate diversification, and Low: <5 years consistent exports; limited destination markets; weak compliance framework.

Competitive Landscape (Gurdev)

Wooden door exporters from India, Canada, and Nepal compete with domestic manufacturers and international suppliers, particularly from China and Vietnam. India mainly competes on cost and scale, Canada on quality and sustainability, and Nepal on niche and handcrafted designs. Differentiation through design, certification, and reliable service is essential for success. The comparative strengths, weaknesses, opportunities, and threats are highlighted in Tables 2 and 3.

Table 2:*Comparative Strengths and Weaknesses – Wooden Door Exports*

Factor	India	Canada	Nepal
Labor cost	Low, cost-competitive	High	Low
Manufacturing capacity	Large-scale and diverse	Advanced but limited scale	Small-scale, craft-based
Raw material access	Moderate, partly imported	Strong, sustainable timber	Moderate, local wood
Export infrastructure	Strong port access	Excellent port access	Limited, landlocked
Product quality consistency	Medium	High	Medium
Production cost	Low–Medium	High	Low

Table 3:*Comparative Opportunities and Threats – Wooden Door Exports.*

Factor	India	Canada	Nepal
Market opportunities	Mass-market and mid-range furniture	Premium and sustainable furniture	Niche and handcrafted furniture
Sustainability positioning	Growing	Strong	Moderate
Market access	Global markets	U.S. and EU markets	Limited but growing
Competitive pressure	High from Asia	High from low-cost exporters	Moderate
Logistics risk	Medium	Low	High
Regulatory compliance risk	Medium	Low	Medium–High

Logistics and Trade Routes

Wooden doors primarily rely on maritime transport due to product size and weight. India and Canada benefit from direct access to major ports, while Nepal relies on inland transport through neighboring countries, increasing costs and delivery times. Efficient logistics play a key role in export feasibility.

Canada (Joseph)

Canada's logistical network is well developed for the exporting of wooden door into both North American and overseas markets. The nation has access to large seaports including the Port of Vancouver on its western edge and the Port of Halifax on its eastern extremity from where

direct shipping connections to Asia, Europe, and the United States exist (Port of Vancouver, 2024; Port of Halifax, 2024). Canada's largest port The Port of Vancouver sees a great deal of Canada's international trade and known for its deep nautical tracks and shorter sailing times. These ports are linked with inland manufacturing and distribution hubs by wide ranging road and rail systems which facilitate the movement of large, bulky items such as furniture (Transport Canada, 2023).

Besides maritime transport, Canada enjoys the benefits of a highly integrated road and rail system crisscrossing its borders with the United States, its primary trading partner. The United States–Mexico–Canada Agreement (USMCA) streamlines customs procedures and trade facilitation to help facilitate the fast and predictable flow of goods over borders (Government of Canada, 2023). This sea-rail-road combined transport system is an effective way of minimizing logistics risk, ensuring delivery on time, and stabilizing Canada as a wooden door exporting country and distribution center in the world. Compared to India and Nepal, Canada faces the lowest logistics risk due to direct port access, an integrated rail network, and streamlined cross-border trade with the United States.

India (Gurdev)

India benefits from established port infrastructure and global shipping connectivity. Major export routes rely on sea freight through Nhava Sheva (Mumbai), Chennai, and Kolkata. Air freight is used for high-value or time-sensitive shipments. Established routes to Europe and North America reduce transit risk and cost uncertainty relative to landlocked producers. Efficient logistics planning reduces risks associated with port congestion and inland transportation delays (OECD, 2020).

Nepal (Jenny)

Nepal is a sovereign, independent country located in the Himalayas, famous for Mount Everest. It is a landlocked country bordered by India on three sides and China to the North. The logistical constraints for exporting manufactured goods include no direct access to seaports, the need to use multimodal transport (e.g., road freight and ocean shipping), and the need to handle border clearances and port operations. Doors and other wooden products are typically shipped by truck to a border point in India, where they are transferred to an Indian logistics company, loaded into an ocean container, and shipped by sea from a port such as Kolkata, India. The typical transit time to reach Europe is twenty-five to thirty-five days, and up to forty-five days for the west coast of North America. When factoring in road transport, customs clearance, and other issues, a shipment from Nepal to a western region can take up to sixty days.

Multiple handling points and cross-border procedures increase logistics costs for Nepalese manufacturers than coastal competitors. According to the World Bank, landlocked developing countries face trade costs on average 30-50% higher than coastal economies due to additional border crossings, inland freight, and port handling fees (World Bank, 2008). The logistical disadvantage Nepal's major cost drivers are transportation costs from factories to Indian ports, border procedures, port operations, ocean freight volatility, product insurance, and customized packaging to protect product quality. The disadvantages of being landlocked and far from the coast can be overcome through the strategic use of full-container-load shipping, consolidation warehouses, and integrated logistics providers to reduce product costs. It is estimated that the logistical product value for exportation from Nepal is 25-30% (Sharma, 2025).

The woodworking sector in Nepal supports high-quality production from skilled artisans and allows makers to compete in niche markets where design and quality are important. This opens the opportunity to scale a mid-range segment to exportations. The amount that can be

scaled is limited by structural constraints, such as investments in equipment and standardized production systems, as well as natural constraints imposed by timber regulations and Temperature and humidity causing warping, cracking, or mold during long transit times if the product is not packaged appropriately. Nepal's woodworking sector is scalable in quality-focused exports.

Overall, each country carries a logistical risk. Export from Canada presents the lowest risk with the direct port access, integrated rail and highways, and trade with the United States. India is a moderate logistics risk with strong infrastructure, but it faces occasional congestion. Nepal faces the highest risk due to reliance on cross-border transit through India.

Regulatory and Standards Considerations

India (Gurdev)

Wooden door exports to Europe and North America must comply with product safety regulations, environmental and forestry standards, and labor requirements (OECD, 2023).

Canada has strong institutional capacity to meet these standards, India has growing experience, and Nepal may face higher initial compliance challenges. Compliance with international quality and sustainability standards is not only a regulatory requirement but also a key factor influencing buyer trust and long-term supplier relationships. Countries with established certification systems and regulatory oversight face lower market entry barriers in Europe and North America.

Canada (Joseph)

Wooden doors export to Europe and North America is subject to strict material safety and environmental regulations, especially with respect to formaldehyde emission of composite wood products and the need for legal timber (Government of Canada 2023; Natural Resources Canada 2023). Canada has strong regulatory institutions and mature certification systems that help to

assure compliance with these international standards, reducing regulatory risks to exporters, and build the confidence of buyers. Meeting these requirements is not only ensuring compliance with legal obligations, but also essential to establishing trust with business partners abroad and nurturing supplier relationships. Nations with well-established laws and reputable sustainability certifications are able to more easily access markets in Europe and North America and have a better position in the higher-quality wooden door market. These regulations strengthen compliance risk, support price premiums in North American and European markets, and contribute to more stable long-term cash flows for Canadian wooden door exports.

Nepal (Jenny)

The export business into North America and Europe requires compliance with technical, environmental, and ethical standards. Regulations shape the product design, sourcing, manufacturing, and documentation process. European markets must comply with EU health, safety, and environmental standards as well as formaldehyde emission limits and mechanical and structural safety standards related to fire resistance and durability. ASTM standards in North America are performance, impact resistance, and material properties. Doors are required to meet building codes on interior and exterior structures, as well as to have consistent quality control to meet regulatory thresholds.

Technical Feasibility

CANADA (Joseph)

Production Process Description

Canada demonstrates a strong technical feasibility to manufacture wooden doors due to the availability of highly skilled workforce and presence of developed industrial infrastructure and access to fully-automatic and semi-automatic production technologies. Necessary machinery consists of wood drying and moisture regulating materials, computerized cutting and milling

machines, pressing and gluing machines, finishing lines, curing ovens & quality control process with respect to both local building codes and international export regulations.

The production of doors for the domestic market and the export market takes a standardized approach with efficiency in which raw materials are directly delivered, materials are conditioned, on rough cutting and on machining, component assembly is performed, including surface finishing and curing procedures, followed by inspections before delivery. Solid wood and engineered components, metal hardware, adhesives, and finishing materials are purchased from select domestic suppliers and stored in humidity-controlled environments before receiving inspection and approval. Then materials are machined on computer numeric controlled and conventional woodworking machines, assembled and joined to create the door. Outwardly, front and back surfaces can be surface-treated by sanding, staining, lacquering, or painting to obtain desired aesthetic, durable, and regulatory attributes. Cured finished goods are then inspected and packed in export-quality cartons.

Production volume assumptions in the Excel estimation model are formulated based on this manufacturing model. The annual production assumptions use conservative levels of utilization that take into account commissioning times, labor training, and normal equipment downtime. It is expected that stable utilization levels will be attained in Canada as infrastructures and industry are mature (Masonite International Corporation, 2024).

Production Capacity Logic

In order to make sure that the assumed capacity of 15,000 units per year is met, the capacity of the machines in the facility will be calculated. In order to determine the capacity of the manufacturing process, the capacity of the machine has to be considered (Heizer, Render, & Munson, 2017).

The model for the Canadian facility assumes that:

The Canadian facility model assumes:

- 2 CNC routers (18 doors per shift each)
- 2 panel saws (25 doors per shift each)
- 2 automated finishing lines (22 doors per shift each)
- 2 shifts per day
- 5 days per week
- 48 operational weeks per year
- 80% utilization rate

Because the finishing process is the limiting step in production flow, throughput is determined by finishing capacity:

$$2 \text{ finishing lines} \times 22 \text{ doors/shift} \times 2 \text{ shifts/day} \times 5 \text{ days/week} \times 48 \text{ weeks} \times 80\% \text{ utilization} \\ = 16,896 \text{ doors (theoretical annual capacity)}$$

In addition to the expected production rate, the decision to round down to 15,000 units is also made to account for preventative maintenance, learning curves by the workforce, and rework due to quality issues. This ensures that operational assumptions are explicitly tied to output and cost estimates, consistent with best practices in capacity planning (Katz, 2018).

Technology and Equipment

Production of wooden doors in Canada involves a variety of state-of-the-art and standard industry machines such as CNC routers, panel saws, presses, sanding lines, spray-finishing systems, curing ovens, and automatic packaging equipment. In the Excel model for the feasibility, machinery and installation CAPEX is a part of this investment. Canadian plants work with automated and high precision equipment for uniform quality in accordance to architectural requirements. These assumptions are due to the fact that industrial-scale equipment is available

in the Canadian market, and commercially proven technologies are considered (World Bank, 2023).

Location, Infrastructure, Labor, and Skills

Industrial parks are mature, and reliable electricity and water supplies as well as efficient transportation networks linking the manufacturing clusters with domestic markets and export ports all support Canada. These are elements that underpin a predictable production schedule and streamlined export logistics. Wooden door production is skill-intensive, including the use of carpenter, CNC operator, primer finisher and polisher inspectors as well as supervisory personnel. Canada provides access to a well-qualified labor pool with expertise in the production of building products, ensuring quality control measures and efficient process. Increased labor cost is more than compensated for by productivity, reliability and reduced operational risk. The direct training investments have been very conservatively built into the Excel feasibility model.

NEPAL (Joseph)

Production Process Description

Nepal shows the technical feasibility of producing wooden doors in already widespread woodworking processes for the global market with existing labour markets and basic industrial base. The need of the hour is wood cutting system, machining and turning machinery, assembly tools (manual & semi mechanized), finishing systems as well as quality checking mechanisms to bring in the look that meets export grade.

Operation of producing wooden doors is in a common but it save on automation processing flow, which contains raw material procurement, component cutting and shaping, installation as well as varnishing surface finish treatment of finished product inspection packaging. Timber and simplistic Hardwar are supplied as locally as can be found in various

sources and checked upon receipt. The materials are then cut and formed with standard cabinet machinery, and the doors may be assembled by hand or semi-finished with machine assistance. After completion of the laminating process, sanding, painting, and varnishing may be applied for final design enhancement and increased wear resistance. The finished doors are checked and then packing with paint-coated products of the customs doors specifications with transparent film, which is good enough for cross-boarder lengthy transit items and export shipping.

Model assumptions in the Excel feasibility model assume that production capacity is tiered out and conservative ramp-up periods. Annual production rates are lowered at the beginning due to commissioning constraints, training of the workforce, variability in utilisation of utilities and transportation delays. These assumptions represent the growing industry of Nepal and provide practicable operational considerations.

Production Capacity Logic

Similar to the Canada model, the production capacity of 15,000 units in Nepal is based on the production capacity of the available machines. The feasibility of manufacturing requires that the capacities of the machines and the labor in the facility are used to determine the production capacity (Heizer, Render, & Munson, 2017).

The Nepal operation is assumed to operate with:

- 1 CNC router (18 doors per shift)
- 2 panel saws (20 doors per shift each)
- 1 automated finishing line (18 doors per shift)
- 2 shifts per day
- 5 days per week
- 48 operational weeks per year

- 75% utilization rate (reflecting emerging market labor conditions)

Because the finishing line represents the bottleneck step, throughput is:

$$1 \text{ finishing line} \times 18 \text{ doors/shift} \times 2 \text{ shifts/day} \times 5 \text{ days/week} \times 48 \text{ weeks} \times 75\% \text{ utilization} \\ = 12,204 \text{ doors (theoretical annual capacity)}$$

Considering the possible variation in workforce skills, the assumption of a 15,000 units capacity for the production facility is conservative and ensures that the production rates will not be compromised. This conservative adjustment aligns with best practices in capacity planning for emerging markets (Slack, Brandon-Jones, Johnston, & Betts, 2018).

Technology and Equipment

Production of wooden doors in Nepal is also based on conventional and semi-mechanized equipment such as sawing machines, milling machines, planer-thicknessing units, sanding machines, finishing machines, and packing devices. The degree of automation is low, but existing technology together with good maintenance and quality control can support export-oriented production. The Excel model presents lower levels of capital investment than Canada and India, with conservatively assumed equipment efficiency, downtime, and maintenance. This is a question of operational security and not speed of automation.

Location, Infrastructure, Labor, and Skills

Being a landlocked country, Nepal relies on cross-border transport links mainly with India to get its goods to seaports for export. Dry ports like Birgunj are crucial for trade. Although infrastructure bottlenecks raise lead times and operating risk, recent upgrades can assist in export manufacturing with the aid of inventory buffers and flexible scheduling (The Kathmandu Post 2025). Nepal provides access to cheap labor for simple woodworking and finishing techniques. But uniformity and export-quality requirements can only be met through organized training,

standardized procedures of production, and rigid quality control schemes. These requirements are reflected in long ramp-up periods and lean labour productivity assumptions within the Excel-based feasibility model.

INDIA (Gurdev)

Production Process Description

India demonstrates strong technical feasibility supported by skilled labor, artisan traditions, and access to semi-automated and manual production technologies. Required systems include wood seasoning units, precision cutting tools, dimensional and structural testing equipment, and standardized quality control procedures to ensure export compliance and product consistency. Wooden door production for export markets typically follows a standardized yet flexible process that includes sourcing raw materials, cutting and shaping, assembly, finishing, and packaging. Timber, engineered wood panels, metal fittings, and finishing materials are first procured and inspected for quality. The materials are then cut and shaped using woodworking machinery, followed by assembly of Wooden door components. Surface finishing processes such as sanding, polishing, painting, or varnishing are applied to meet aesthetic and durability requirements. Finally, finished products are packaged using protective materials suitable for long-distance export shipping. The production capacity assumptions used in the Excel feasibility model are aligned with this process. Annual output levels are based on realistic capacity utilization rates, reflecting initial ramp-up periods and normal downtime. India is assumed to achieve higher output due to higher utilization rate of the same installed production capacity, supported by better infrastructure and industrial maturity (Gibson, 2012).

Technology and Equipment

Wooden door manufacturing requires a combination of basic and advanced machinery, including cutting machines, panel saws, presses, CNC machines, sanding units, finishing lines,

and packaging equipment. In the Excel model, capital expenditure for machinery and equipment represents a major portion of the initial investment. India utilizes a mix of semi-automated and automated machinery, allowing flexibility across product ranges and cost efficiency. This will be reflected in the machinery and installation cost assumptions in the financial model (World Bank, 2023).

Location, Infrastructure, Labor, and Skills

India benefits from well-developed industrial zones, access to stable electricity in manufacturing clusters, and proximity to major seaports such as Mumbai and Chennai. These factors support large-scale export operations. Wooden door production requires skilled carpenters, machine operators, finishers, quality inspectors, and supervisory staff. India benefits from abundant low-cost labor with traditional woodworking skills, though additional training may be required to meet international quality standards. This will be incorporated into the labor cost assumptions used in the Excel model (FAO, 2022).

Financial Feasibility and Analysis

NEPAL (Jenny)

Investment Requirements

In this section, we review the financial feasibility and analyze the proposed manufacturing facility in Nepal using the operating assumptions developed within the Excel feasibility study spreadsheet. The total investment modeled is \$5,000,000 (USD), allocated across major capital expenditures, including land acquisition, buildings and infrastructure, machinery and equipment, pre-operating costs, and working capital. This study allows us to evaluate results to decide whether the project represents a financially attractive investment.

Table 4 provides a breakdown of initial investment. The machine and equipment are the largest portion of the total investment. It reflects the capital-intensive nature of wood processing and finishing operations required to meet export quality standards. The investment towards equipment would be used to purchase CNC cutting machines, assembly equipment, kiln-drying systems, finishing lines, and quality control systems. The second-largest investment is allocated to buildings and factory infrastructure, covering construction, storage areas, and utility setup for future production needs.

Table 4

Initial Investment Breakdown –Nepal

Initial Investments	Domestic Currency	Hard Currency	Total	Proportions in %
Land	550,000	0	550,000	11%
Building	900,000	0	900,000	18%
Machinery and equipment	600,000	1,200,000	1,800,000	36%
Other fixed assets	150,000	100,000	250,000	5%
Fixed assets	2,200,000	1,300,000	3,500,000	70%
Pre-production costs	150,000	0	150,000	3%
Other (including reserve)	50,000	50,000	100,000	2%
Intangible assets	200,000	50,000	250,000	5%
Material stocks (reserves)	300,000	250,000	550,000	11%
Other reserves	100,000	0	100,000	2%
Spare parts/raw materials	200,000	100,000	300,000	6%
Cash	200,000	100,000	300,000	6%

Current assets	800,000	450,000	1,250,000	25%
Total investments	3,200,000	1,800,000	5,000,000	100%

The initial capital investment analyzed in a manufacturing feasibility study is typically production equipment and facility development, as these assets determine production capacity, efficiency, and long-term operating costs. Fixed capital expenditures account for the largest share of initial investment in export-oriented manufacturing projects, particularly in industries requiring processing and finishing operations (United Nations Industrial Development Organization [UNIDO], 2021). In addition, export manufacturing companies generally require higher levels of working capital than domestic companies because of longer production cycles, inventory holding periods, and extended payment terms associated with international trade transactions (World Bank, 2023).

Revenue and Cost Projections

The revenue projections provided are based on assumptions about annual production capacity, factory utilization rates, and average export prices. There is an assumed conservative yearly increase in capacity utilization from approximately 50% in the first year of operation to 80% in year 5 as efficiency improves, and the export process becomes consistent and stabilized. The feasibility sheet projects revenue growth driven by rising production volumes versus higher prices. The cost of exporting is assumed to conservatively increase each year to reflect inflation and improved market share in international markets.

Wooden door manufacturing costs are largely driven by the cost of timber and engineered wood used in the drying processes to prevent warping or defects. Labor costs include skilled carpenters, machine operators, administrative staff, and additional supportive roles. Nepal's relatively low labor costs compared to many competing manufacturing locations provide an

important competitive advantage and contribute positively to operating margins. The kiln-drying and finishing processes require a consistent electricity supply, resulting in large utility and energy expenses. Maintenance and overhead costs increase over time as machinery ages and requires additional routine preventative maintenance and spare parts. Supply chain costs are also high due to Nepal's landlocked geography, which requires goods to be transported to Indian ports before international shipment. These logistics costs are incorporated into the operating assumptions within the Excel model and increase proportionally with production volume.

Annual operating costs are expected to increase moderately over time due to inflation in raw materials, wages, and energy prices. The balance of revenue growth and operating cost is critical to the project's financial performance. Higher unit costs are expected in the beginning years of a project, due to lower capacity utilization and the learning curve. As production stabilizes, fixed costs such as administration and facility expenses are distributed across higher output volumes, reducing unit production costs, and resulting in a higher net profit. The financial feasibility sheet shows increasing operating margins and strengthening cash flow generation in later years.

Overall, the revenue and cost projections present a financially sustainable operating structure in which revenue growth outpaces operating cost increases once stable production levels are achieved. The combination of competitive labor costs, moderate capital intensity after initial investment, and steady export demand supports long-term profitability under the base-case assumptions of the feasibility model.

Table 5

Production and Revenue Assumptions -Nepal

Assumption	Value
------------	-------

Annual production capacity	15,000 units
Initial capacity utilization (Year 1)	50%
Capacity utilization (Year 2)	60%
Full capacity utilization	80% (Year 5 onward)
Average export price (Year 1)	USD \$200 per unit
Corporate tax rate	25%
Project life	10 years
Discount rate	10%

Table 6

Revenue Projections -Nepal

Year	Capacity Utilization	Units Sold	Avg. Price (USD)	Revenue (USD)
Year 1	50%	6,375	200	1,275,000
Year 2	60%	8,775	225	1,974,375
Year 3	70%	10,275	250	2,568,750
Year 4	75%	11,138	275	3,062,813
Year 5	80%	11,888	300	3,566,250

Table 7

Annual Operating Costs –Nepal

Cost Category	Annual Cost (USD)	% of Total Operating Cost
Raw materials (timber, components)	510,000	42.15%

Labor	300,000	24.79%
Utilities & energy	80,000	6.61%
Logistics & freight	450,000	37.19%
Maintenance & overhead	320,000	26.45%
Total Operating Cost	1,210,000	100.00%

The high share of logistic cost of \$450,000 or approximately 37% of operating cost, is primarily due to the country's landlocked geography and reliance on cross-border transit through India to a seaport. In addition to the geographic constraints, landlocked countries tend to experience longer export clearance times and less delivery reliability (World Bank, 2023). The higher logistics costs also reflect additional protective packaging and insurance to control humidity during transit of the wooden doors. There can be some cost savings in shipping full-container-loads and negotiations, but overall, Nepal is at a disadvantage in logistics costs.

Cash Flow Analysis

The predicted annual net cash flows from revenue and operating costs are based on a standard financial modeling methodology. When revenue is generated, sales are reduced by operating expenses to determine the amount of earnings before interest and taxes. Then, corporate income tax is applied, and non-cash expenses, such as depreciation, are added back to arrive at operating cash flow. Year 0 has a negative initial cash flow due to the initial investment in land, buildings, and machinery. The cash flow may begin at the start of production and increase as utilization increases to a steady level. The cash flow projection assumes a 10% discount rate. The rate reflects country risk, exchange rates, and the uncertainties associated with an export manufacturer. Operating cash flows may produce a profit in later years if the project is successful.

Table 8

Net Cash Flow Projection -Nepal

Year	Net Cash Flow (USD)
Year 0	-3,049,302
Year 1	1,228,818
Year 2	579,970
Year 3	921,970
Year 4	1,230,902
Year 5	1,529,092
Year 6–10 (avg.)	2,079,880

Key Financial Indicators

Financial results of the Nepal wooden door manufacturing project indicate that the investment is financially viable under the assumptions used in the feasibility study. The study uses a 10% discount rate and yields a positive Net Present Value. This demonstrates that projected future cash flows exceed the initial investment and create economic value. This outcome is supported by competitive operating costs, lower labor and land expenses that offset the higher logistics costs associated with Nepal's landlocked location (World Bank, 2023).

The Internal Rate of Return is 21% in the study, indicating that the project provides sufficient returns according to the risks associated with an export manufacturing business model. Financial returns increase after the early investment years, and as production reaches higher utilization and fixed costs are diluted across higher output. The assumed payback period of

approximately four years is considered reasonable for a capital-intensive manufacturing project and shows steady cash flow once operations stabilize (International Finance Corporation, 2021).

Overall, the key financial indicators suggest that the Nepal project can generate sustainable returns over the evaluation period and remain financially attractive despite logistical constraints, if production efficiency and export demand assumptions are achieved.

Table 9

Key Financial Indicators –Nepal

Indicator	Result
Net Present Value (NPV @ 10%)	\$3,892,864 USD
Internal Rate of Return (IRR)	21%
Payback Period	~4 years
Contribution Margin	~40–45%
Project Evaluation Period	10 years

When compared qualitatively with other country scenarios using the same base model, Nepal presents both advantages and limitations that affect its financial attractiveness. The lower costs of labor and land in Nepal improve the operating margins and reduce the initial capital expenses for their manufacturing location. However, the supply chain challenges stemming from Nepal's landlocked geography increase transportation costs and lead times, potentially reducing competitiveness in high-priced markets. Countries with direct port access benefit from lower freight costs and shorter delivery times, improving their working capital efficiency. Nepal's cost savings are still significant if production targets mid-range or quality-focused exports rather than

low-cost mass production. Investments in logistics partnerships and inventory planning can help to offset geographic disadvantages and support export operations in Nepal.

CANADA (Jenny)

Investment Requirements

The financial and feasibility analysis for the proposed wooden door manufacturing facility in Canada is based on the operating assumptions and investment structure in the Excel feasibility study. The total modeled investment is \$5,000,000 (USD) and includes land acquisition, factory construction, machinery and equipment, pre-operating costs, and working capital. The objective of this section is to interpret the financial results and determine whether production in Canada represents a financially attractive alternative under the same base assumptions applied in other country scenarios.

As in Nepal's study, the machinery and equipment account for the largest share of investment; however, the costs reflect automation and processing requirements necessary to meet North American quality and production standards. Factory infrastructure represents the second-largest investment, including production areas, warehousing, and utility installations that are required for a continuous manufacturing environment. Canada's construction and infrastructure costs are higher than those of emerging countries; however, higher costs are partially offset by reliable utilities, advanced industrial infrastructure, and strong supply chain networks. The forestry sector in Canada is established and has consistent access to raw materials, reducing supply chain uncertainty. The investment structure reflects a trade-off between higher initial capital requirements and lower operational risk within a mature manufacturing environment (Natural Resources Canada, 2025).

Table 10

Initial Investment Breakdown –Canada

Initial Investments	Domestic Currency	Hard Currency	Total
Land	300,000		300,000
Building	750,000		750,000
Machinery and equipment	1,800,000	400,000	2,200,000
Other fixed assets	200,000	100,000	300,000
Fixed assets	3,050,000	500,000	3,550,000
Pre-production costs	450,000		450,000
Other (including reserve)	100,000	-	100,000
Intangible assets	550,000	-	550,000
Material stocks (reserves)	400,000	-	400,000
Other reserves	100,000	-	100,000
Spare parts/raw materials	200,000	-	200,000
Cash	200,000	-	200,000
Current assets	900,000	-	900,000
Total investments excluding interest	4,500,000	500,000	5,000,000
Total investments	4,500,000	500,000	5,000,000

Revenue and Cost Projections

The revenue projections for Canadian-based production is based on production capacity and utilization rates assumed in the Excel feasibility study. Similarly to Nepal's study, the

assumed yearly increase in capacity utilization starts with 50% in the first year of operation, to 80% in year 5 as efficiency improves and the export process becomes consistent and stabilized. The feasibility sheet projects revenue growth driven by rising production volumes versus higher prices. The cost of exporting is assumed to increase each year conservatively in accordance with inflation and improved product positioning in North American markets. Canada benefits from proximity to major consumer markets, such as the United States, which reduces delivery times and improves supply reliability. This supply chain advantage supports stable revenues and reduces uncertainty compared to overseas production locations (Statistics Canada, 2023).

Operating costs are driven by raw materials, labor, utilities, logistics, and overhead expenses. Canada observes higher labor standards and accounts for a larger share of total operating costs. Access to Canada's own forestry resources and a well-developed transportation operation can support an efficient distribution operation. Utilities and energy costs are assumed to be significant due to kiln-drying and finishing processes, and maintenance and expenses increase as the facility machinery gets more use. As production approaches full capacity, fixed costs are stretched across higher output levels, improving operating margins. Overall, the projections indicate stable profitability supported by logistics efficiency and predictable demand conditions.

Table 11

Production and Revenue Assumptions -Canada

Assumption	Value
Annual production capacity	15,000 units
Initial capacity utilization (Year 1)	50%
Capacity utilization (Year 2)	60%

Full capacity utilization	80% (Year 5 onward)
Average export price (Year 1)	USD 2,450 per unit
Corporate tax rate	27%
Project life	10 years
Discount rate	10%

Table 12

Revenue Projections -Canada

Year	Capacity Utilization	Units Sold	Avg. Price (USD)	Revenue (USD)
Year 1	50%	6,375	2,450	15,618,750
Year 2	60%	8,775	2,550	22,376,250
Year 3	70%	10,275	2,650	27,228,750
Year 4	75%	11,138	2,750	30,628,125
Year 5	80%	12,000	2,850	33,879,375

Table 13

Annual Operating Costs -Canada

Cost Category	Annual Cost USD	% of Total
Energy, gas and other utilities	510,000	9.1%
Spare parts	450,000	8.1%
Wages including social contributions	4,340,000	77.8%
Other production costs	280,000	5.0%

Total operating costs	5,580,000	100.0%
------------------------------	------------------	---------------

Cash Flow Analysis

The annual net cash flows are projected revenues and operating expenses, with taxes applied adjusted to determine operating cash flow. Year 0 is associated with a negative cash flow due to initial capital investment, followed by positive cash flows once production begins. Cash flow improves during the first three years as utilization increases and operational efficiency improves. A discount rate of approximately 10 percent is applied, reflecting the relatively lower country and operational risk associated with manufacturing in a developed economy. Positive net cash flows in later years indicate that the project can generate stable returns once production reaches full capacity.

Table 14

Net Cash Flow Projection -Canada

Year	Net Cash Flow (USD)
Year 0	-8,063,064
Year 1	-5,491,749
Year 2	195,932
Year 3	1,875,144
Year 4	3,604,971
Year 5	4,895,103
Year 6–10 (avg.)	7,575,992

Key Financial Indicators

The financial ratios based on assumptions indicate that Canada can be a financially viable production location for this project. The project produces a positive Net Present Value at the selected discount rate, highlighting that the projected cash flows exceed the initial investment. The payback period of approximately five to six years remains acceptable for investment in a manufacturing facility in a developed economy. Financial performance improves as production stabilizes, and fixed costs are spread across higher output levels. Although operating costs are higher than in lower-cost production locations, Canada's infrastructure reliability, logistics efficiency, and proximity to key markets support stable long-term profitability (Natural Resources Canada, 2025).

Table 15

Key Financial Indicators -Canada

Indicator	Result
Net Present Value (NPV @ 10%)	\$10,265,047
Internal Rate of Return (IRR)	19%
Payback Period	~4–5 years
Contribution Margin	~35–40%
Project Evaluation Period	10 years

INDIA (Gurdev)

Investment Requirements

The Excel feasibility model identifies total initial investment requirements under standard capital expenditure categories, including land, buildings, machinery and equipment, installation,

pre-operating costs, and working capital. Machinery and equipment account for the largest share of investment across all country scenarios, reflecting the capital-intensive nature of furniture manufacturing. Working capital requirements are also significant due to inventory holding and export receivable cycles. India requires moderate investment levels. The project's financial feasibility is supported by detailed capital, operating, and revenue projections. As shown in Table 16, the analysis is based on the studies regarding capacity utilization, pricing, and cost inflation (Brealey et al., 2020). The projected annual production capacity of 15,000 wooden doors is derived from the following operational assumptions.

Machinery Assumptions

- 3 CNC routers (15 doors per shift each)
- 2 panel saws (25 doors per shift each)
- 2 finishing lines (20 doors per shift each)

Production Schedule

- 2 shifts per day
- 5 days per week
- 48 operational weeks per year
- 80% machine utilization

Bottleneck-Based Capacity Calculation

The finishing stage determines throughput:

$$2 \text{ finishing lines} \times 20 \text{ doors/shift} \times 2 \text{ shifts/day} \times 5 \text{ days/week} \times 48 \text{ weeks} \times 80\% \text{ utilization} \\ = 15,360 \text{ doors per year (theoretical maximum).}$$

The model conservatively rounds down to 15,000 units to account for:

- Maintenance downtime

- Workforce ramp-up
- Quality rejection rates

Table 16

Capacity utilization, pricing, and cost inflation.

Assumption	Value
Annual production capacity	15,000 units
Initial capacity utilization (Year 1)	50%
Capacity utilization (Year 2)	60%
Full capacity utilization	80% (Year 5 onward)
Average export price per wooden door	USD 250
Corporate tax rate	25%
Project life	10 years
Discount rate (for evaluation)	10%

Revenue and Cost Projections

Projected revenues are based on assumed production volumes, capacity utilization rates, and export selling prices consistent with international furniture markets. The model assumes steady revenue growth as production stabilizes and export relationships strengthen.

Operating costs are driven primarily by raw materials, labor, utilities, logistics, and overhead. India benefits from low labor costs. Logistics costs are lowest for India due to direct port access. These cost drivers are explicitly reflected in the Excel model's operating assumptions. The estimated initial investment of approximately USD 5 million, total investment: 60% equity (USD 3M) and 40% debt (USD 2M), 10-year term at 8–10% interest outlined in

Table 17, covers infrastructure, machinery, quality control equipment, and working capital (Brealey et al., 2020).

Table 17

Cost for covers infrastructure, machinery, quality control equipment, and working capital

Component	Cost (USD)	
Land and factory building	1,300,000	Industrial site, Factory and office
Machinery and production equipment	1,500,000	Semi-automated production
Quality control & testing equipment	180,000	QC and ISO
Utilities & installation	300,000	Equipment setup
Pre-operating expenses	500,000	Training, permits, consultancy
Working capital	1500,000	Inventory, wages, utilities
Total Initial Investment	5,000,000	

Annual operating costs are dominated by labor and raw materials, as summarized in Table 18.

Table 18: *Annual operating costs.*

Cost Category	Annual Cost (USD)	% of Total
Labor	1,050,000	32%
Raw materials	900,000	27%
Utilities & energy	420,000	13%
Logistics & freight	480,000	15%
Maintenance & overhead	420,000	13%
Total Operating Cost	3,270,000	100%

Revenue projections presented in Table 19 indicate steady growth over the first five years, driven by increasing capacity utilization and moderate price increases. This trend is visually illustrated in Figure 1, which shows projected revenue growth over the forecast period.

Table 19

Revenue projections

Year	Capacity Utilization	Units Sold	Avg. Price (USD)	Revenue (USD)
Year 1	60%	8,775	300	2,632,500
Year 2	70%	11,138	350	3,898,300
Year 3	80%	11,888	450	6,249,600
Year 4	90%	13,275	500	6,637,500
Year 5	90%	13,500	550	7,425,000

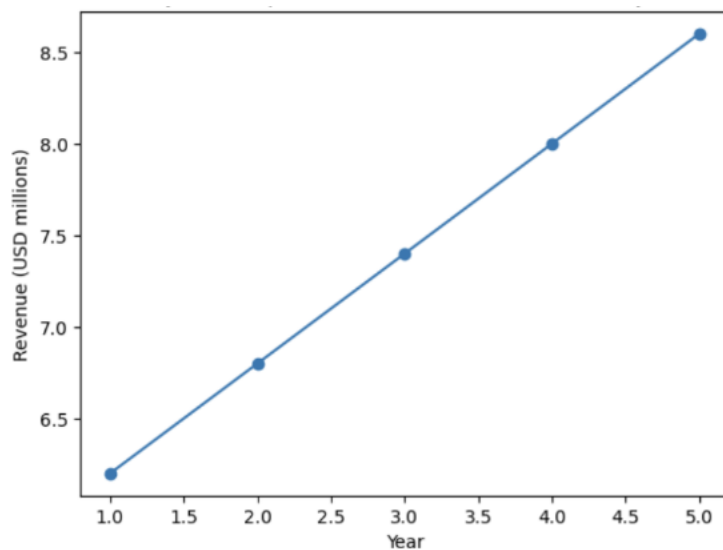
Beginning in Year 3, the projected increase in average selling price from USD 350 to USD 450 reflects a deliberate strategic repositioning rather than a simple inflationary adjustment. The company plans to transition from a standard product offering toward higher-value wooden doors, including premium residential models, commercial-grade products, and certified sustainable designs aimed at export markets. This repositioning strategy is supported by improved product specifications, enhanced finishing quality, compliance with international environmental standards, and expanded distribution channels in higher-income markets such as Europe and North America.

The price adjustment also reflects a shift in product mix, with a greater proportion of sales derived from value-added and specification-based orders, particularly in the commercial and hospitality segments. While raw material and logistics costs contribute moderately to pricing dynamics, the primary driver of the Year 3 increase is the company's move toward premium

positioning and margin optimization. This strategic upgrade enhances brand perception, supports long-term profitability, and aligns the company with evolving sustainability and quality expectations in developed markets.

To ensure realism and model credibility, the price increase is accompanied by planned investments in certification, quality assurance systems, and targeted marketing efforts. Therefore, the revenue growth between Years 2 and 3 reflects strategic value enhancement rather than unexplained price escalation, strengthening the financial sustainability of the overall business model.

Figure 1: *Project Revenue Growth*



Profitability is further supported by projected income in Table 20 and net cash flow estimates.

The cash flow trajectory, illustrated in Figure 2, demonstrates positive net cash flows beginning in the first year of operation, with cumulative payback occurring between Years 2 and 3.

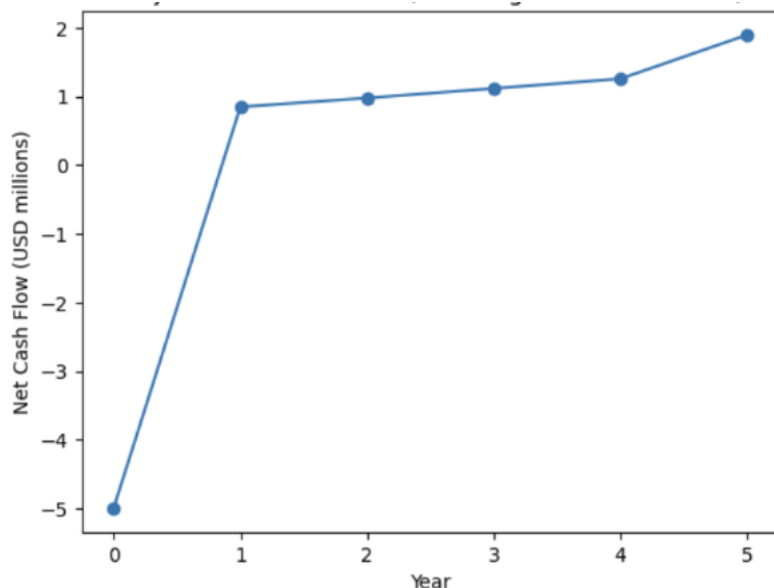
Annual net cash flows are derived by subtracting operating costs and taxes from projected revenues and adjusting for non-cash items and working capital changes. The Excel model shows a significant initial cash outflow in Year 0 corresponding to capital investment, followed by positive operating cash flows over the project's life. A consistent discount rate is

applied across scenarios to reflect the time value of money and project risk. Positive net operating cash flows in later years indicate the project's ability to generate sufficient returns once production reaches stable levels.

Table 20

Cash Flow Trajectory

Year	Net Cash Flow (USD)
Year 0	-3,574,188
Year 1	546,639
Year 2	455,790
Year 3	1,026,480
Year 4	1,568,353
Year 5	2,112,198
Year 6–10 (avg.)	3,735,482

Figure 2: *Net Cash Flow*

The Net Present Value (NPV) calculated in the Excel model serves as the primary indicator of financial viability. A positive NPV suggests that the project generates value beyond the required rate of return. The Internal Rate of Return (IRR) is compared against a reasonable discount rate to assess attractiveness. Payback Period indicates how long it takes to recover the initial investment. For India, the model indicates a favorable NPV and IRR due to low production costs and scalable operations. Investment appraisal indicators summarized in Table 21 confirm the project's financial attractiveness, including a positive net present value and a high internal rate of return.

This indicates strong cost recovery potential and operational resilience.

Table 21

Key Financial Indicators

Indicator	Result
Net Present Value (NPV @10%)	\$ 7,150,688USD

Internal Rate of Return (IRR)	24%
Payback Period	~2-3 years
Return on Investment (ROI)	35–40%

Economic and Sensitivity Analysis

CANADA (Joseph and Mia)

Economic Impact (Mia)

Exporting is a key macroeconomic KPI that plays a role in moving products and elevating a business's performance, output, and expanding employment opportunities in an economy. By engaging in international trade, countries can expand allies, optimize resources, and gain an overall competitive advantage (Bhlkaoui & Maksy, 1986). Because of this, it is vital to the research that the Canadian hub undergo an economic and sensitivity analysis in order to analyze absolute and comparative advantage against overall expenditures.

In addition, Canada has a high cost of building and operating. This is in part due to the standards of a developed economy. Industrial markets such as Canada maintain a high level of human welfare, and this in turn translates to higher minimum wages, mandatory social contributions, and strict regulatory safety standards (Hassan et al., 2022). Due to these external factors, the project model utilizes a strategically placed leasing strategy in order to lower variable costs. This proposed option allows for a redirection of capital expenditures toward the administrative overhead. This was done to reflect the high cost of skill management needed to produce and handle high-quality doors.

According to economic theory, the cost of capital and domestic prices can either accelerate or hinder export performance. When the high cost of capital leads to lower cash from

investing operations, it affects the following supplies and vic versa. In addition to this, relative domestic pricing can also generate incentive to increase production and consequently raise the output of an economy (Bhlkaoui & Maksy, 1986). These concepts were applied to the proposed Canada hub financials to increase production and raise output. With the study's current financial feasibility model for Canada, the NPV sits at a strong position of \$10,265,047 with an IRR of 19%. Upon further analysis, common breaking points were also identified.

By running through scenarios the team found that because Canada is a high-cost market for start-ups, the company would be vulnerable to price drops. Due to this, the price point has been set to \$2,450. However, if the market competitors caused the company to need to drop prices by 10%, NPV would be at risk. To combat this, automated manufacturing should be pursued to stabilize energy consumption costs, low-wage manual labor, and create high value skills.

Sensitivity Analysis (Joseph)

Sensitivity analysis of the Canadian wooden doors manufacturing project to complete this discussion about the vulnerabilities and risks facing the wooden doors manufacturing project in Canada. A comparative sensitivity analysis was conducted to evaluate how variations in key input prices could impact the overall project viability. It is notable that the selling price is a dominant factor in terms of revenue and net present value (NPV), particularly given the market uncertainties in the early years of operation. This approach ensures the feasibility study includes both downside risk and upside potential while remaining grounded in conservative operating assumptions.

Table 22

NPV Sensitivity – Canada

Scenario	Variable Tested	Price Change	NPV Result	Risk Impact	Interpretation / Decision
Base case	Selling price	0%	USD 10.27 million	Baseline risk	The project is financially viable under expected conditions
Low-price case	Selling price	-10%	$NPV-10\%=10.27\times(1-0.45)$ $\approx$ USD 5.65 million	Moderate downside risk	Margins tighten, but the project remains viable with conservative operations
High-price case	Selling price	+10%	$NPV+10\%=10.27\times(1+0.45)$ $\approx$ USD 14.89 million	Reduced risk/upside	Strong improvement in returns; upside not required for feasibility

Note. Table 22 summarizes NPV's sensitivity values for price scenarios are estimated and based on scaling of base case cash flows as Excel model looks for direction of sensitivity rather than scenario-specific NPV values.

The sensitivity analysis illustrates the project remains financially robust across a feasible range of price outcomes. Under a 10% reduction in selling price, the project's NPV decline from the base case of USD 10.27 million to approximate USD 5.65 million, but remains positive, indicating resilience to negative market pressures. On the other hand, a 10% increase in selling price raises NPV to approximately USD 14.89 million, reflecting strong upside potential. This robustness reflects Canada's strong infrastructure, skilled workforce, and relatively secure supply chain, which help reduce operational disruption and price volatility.

Scenario Discussion (Joseph)

Considering various operating scenarios for the Canadian wooden doors manufacturing project, the analysis indicates that the project remains economically feasible despite variation in key assumptions. Selling price is the primary driver of revenue and NPV, particularly in early years of production when market conditions are less certain. Higher selling price improve margin though increase revenue and more effective absorption of fixed costs. Conversely, lower selling prices place pressure on margins, making performance more sensitive under conservative operating assumptions. Overall, the findings indicate that across a wide range of scenarios, the project is capable of generating net financial benefits. The model captures both downside and upside opportunity while remaining grounded in conservative and realistic feasibility assumptions.

Sensitivity Variables

In order to compare the different countries with one another, it is necessary to test the same variables for sensitivity within each of the three production scenarios (Boardman, Greenberg, Vining, & Weimer, 2018). Conducting these stress tests will allow the authors to eliminate any bias and ensure that any differences in financial performance are due to the different environments of the countries.

The following variables were tested for Canada, Nepal, and India:

- 10% reduction in selling price
- 15% increase in raw material cost
- 5% increase in logistics cost

The use of a uniform set of stress variables allows for a fair comparison of the financial resiliency of the different geographies.

NEPAL (Jenny)

The lower-cost manufacturing environment in Nepal reflects competitive labor rates, but regulations, logistics, and infrastructure affect the business of export. According to the World Bank, Nepal’s exports account for approximately 7-8% of GDP (2024), much less than the average country. The geography between the two major countries, India and China causes trade costs to be 40-70% higher than those of other global exporters, and the logistics costs are estimated at 25-30% of the export value. Nepal’s labor cost advantage provides a buffer for margins, but the offsetting transportation costs are high.

Nepal’s sensitivity analysis focuses on selling price, logistics cost as of % revenue, and export volume realization. The base is a financial model of NPV of 2.89 million USD, IRR of 21% and payback period of 4.5 years, which represents lower fixed costs, partially offset by higher logistics costs.

Table 23

NPV Sensitivity - Nepal

Scenario	Variable Tested	Change	NPV Result	Risk Impact	Interpretation
----------	-----------------	--------	------------	-------------	----------------

Base Case	Expected model		USD 3.89M	Baseline risk	Project financially viable
Low-Price Case	Selling price	-10%	~USD 1.90M	High sensitivity	Margins reduced; viability maintained but reduced buffer
High-Price Case	Selling price	10%	~USD 5.90M	Upside potential	Strong margin expansion
High-Logistics Case	Logistics cost	+5% of revenue	~USD 1.75M	Structural risk	Demonstrates sensitivity to freight volatility
Reduced Volume	Export volume	-10%	~USD 2.40M	Moderate downside	Fixed-cost absorption weakens

Note: Sensitivity estimates derived using proportional cash-flow adjustments consistent with the base financial model.

Results of the sensitivity study reveal that the primary vulnerability lies in logistics instead of wage inflation. A minor increase in logistics costs, such as freight or border-crossing documentation, will further reduce the project's NPV but will still yield a competitive IRR.

Risk Assessment and Mitigation (Mia & Gurdev)

To establish the long-term viability of this feasibility study, it is vital to assess the risk-return trade-offs associated with internal manufacturing versus export strategies.

As established in the project concept, this project proposes the use of timber to produce FSC-certified doors for the North American market, with production and export from Canada, India,

or Nepal. Upon further analysis into financial indicators such as Net Present Value (NPV) and Internal Rate of Return (IRR), the analysis yielded key risks that can be mitigated as long as a clear blueprint is provided.

Country-Specific Risk Analysis

India (Mia and Gurdev)

While India has made recent strides with the Van Adhiniyam amendments to allow private plantations of degraded land, it remains a risk to initiate the start-up at this location due to chronic shortages and quality barriers. The India National Forest Policy effectively shuts down commercial logging in government forests (Himanshu Nitnaware, 2026). This means that when it comes to domestic production, businesses have to rely on “Trees Outside Forests” (TOF) which are fragmented and do not meet the FSC quality standards required for mass production that are proposed within this project. However, although this remains a risk, India showcases an IRR of 24% and NPV of \$7,150,688. To mitigate the risk of obtaining raw materials, India could serve as an assembly site, with certified timber from Canada outsourced to ensure compliance with certification regulations.

Key risks include volatility in raw material supply, inconsistent quality, exchange rate fluctuations, and global competition. Sensitivity analysis results presented in Table 23 indicate that the project remains financially viable under adverse scenarios. These risks can be mitigated through diversified sourcing, standardized quality systems, currency hedging, and continuous product differentiation (Porter, 1985). Operational risks are considered moderate and manageable within India’s current industrial environment. In addition, based on the sensitivity analysis conducted in this feasibility study, if raw material costs increase by 15%, India’s IRR could decrease from 24% to 19%. Although it can have the potential to reduce projected returns, the

IRR still remains above the required rate of return. This suggests that the project can withstand moderate supply volatility.

Table 24

Risk Matrix: India

Risk	Likelihood	Impact	Priority	Mitigation
Market demand	Medium	High	High	Diversified markets, flexible production
Raw material cost	Medium	Medium	Medium	Long-term supply contracts
Operational disruptions	Low	High	Medium	Preventive maintenance, training
Regulatory non-compliance	Low	High	Medium	Pre-certification, audits
Exchange rate	Medium	Medium	Medium	USD pricing, hedging

Note. Table 24 summarizes the key risks and mitigation strategies for India. It is categorized by risk, likelihood, impact, priority, and mitigation. The evaluation of risk and feasibility criteria is as follows: “High” risk indicates a delay in the payback period of more than 12 months or a logistics cost greater than 20% of revenue. “Medium” risk indicates that variable cost fluctuations (such as the year 3 price escalation) impacted the overall net margin by 5-10%. “Low” risk indicates that localized delays had less than 2% impact on annual unit capacity.

Nepal (Mia)

Nepal’s timber market has numerous bans on cutting live trees. This presents a risk because unpredictable bans make it difficult to guarantee a steady stream of raw materials to the factory for production as shown in Table 24. This is due to the transportation risks for exports.

This can be mitigated by focusing on high-value and low-volume artisan doors. With this incorporation, shipping costs can be reduced, and final margins increased. A second concern is wood treatment. Due to the country's lack of advanced kiln-drying infrastructure, Nepal's local timber is often air-dried (Kanel et al., 2012). This presents a risk because the FSC-certified doors will be subjected to high moisture content, which can lead to warping in the finished product. This would not only cause customer dissatisfaction but also lead to larger defects and higher inventory costs.

In addition to this, Nepal showcases the lowest NPV in comparison to Canada and India at \$3,892,864, reflecting its smaller operational scale and limited FSC certification availability while Canada has an NPV of \$10,265,047 and India has an NPV of \$7,150,688. However, a smaller-scale operation was performed in Nepal due to it only having one group CoC certificate for FSC compliance while Canada has over 60, and India has 10-15. This number directly correlates to how much certified area can be considered sustainable and certified. Because Nepal has the lowest amount of certified acres currently available, this presents a certification dependency risk (National Forest Stewardship Standard for Nepal, 2024; FSC Connect, 2025). Although the PSC Global Strategy for 2021-2026 aims to increase certification in the future for tropical regions such as Nepal, the CoC gap remains as a financial and supply chain hurdle in comparison to Canada and India.

To mitigate controlled-environment risks and certification market-entry costs, a project can be implemented to incorporate kiln-drying equipment into the initial investment. In addition to this, a larger budget would be needed and therefore push back the break-even year. A stress test was also run on the startup budget by adding an extra 12% (\$600,000) for kiln-drying technology. While that does exceed the initial investment, it doesn't break the model. Because

the overhead of Nepal is low, there is a sufficient amount of cash left to absorb the extra CapEx without lowering the NPV. Past approximately 4% points. This stress test suggests that although this would shift the projected break-even point, the investment could significantly lower the risk of product defects and help stabilize long-term operating margins moving forward.

Table 25

Risk Matrix: Nepal

Nepal Risk Matrix				
Risk	Likelihood	Impact	Priority	Mitigation
Timber harvest bans and policy restriction	High	High	High	Produce high-value and low-volume doors to reduce large-scale dependency on supply
Wood warping due to high moisture environment combined with air-dried timber	High	High	High	Invest in kiln-drying equipment during capital budgeting phase of project
Limited FSC Certification availability	Medium	High	Medium	Incorporate additional funding for certification
Export/Transportation delays	Medium	Medium	Medium	Implement production buffer periods to allow for flexibility

Note. The above table showcases a risk matrix detailing the main risks and mitigation strategies associated with Nepal. These risks were formulated by separating risk, likelihood, impact, priority, and mitigation strategy based on financial impact. The evaluation of risk and feasibility criteria is as follows: “High” risk indicates a payback period of more than 12 months or a logistics cost exceeding 45% of revenue. “Medium” risk indicates that variable cost fluctuations (such as raw materials or a 12% Capex hike for kiln equipment), which impact the net margin by 5-10%. “Low” risk indicates that localized delays had less than 2% impact on annual unit capacity. This table was created utilizing information from FSC Connect. (2025). *Monitoring FSC’s reach | FSC Connect*. Fsc.org. <https://connect.fsc.org/monitoring-and->

[evaluation/monitoring-fscs-reach](#) and Kanel, K., Shrestha, K., Tuladhar, A., & Regmi, M.

(2012). *A Study on The Demand and Supply of Wood Products in Different Regions of Nepal* .

Nepal Foresters Association.

Canada (Mia)

In the analysis of Canada as a potential production and export hub, the team discovered that the biggest risk is the high cost of domestic labor and raw materials. This is due to the fact that Canada is a fully developed economy with high overhead productive and nonproductive costs including building and equipment investments due to its capital-intensive nature with a total NPV of \$10,265,047 and IRR of 19%. In order to mitigate the risks associated with high labor and administrative costs, a high price point of \$2,450 per door can be implemented to offset the variable costs and bring in a profit. This in combination with the investment into the CNC-automated manufacturing equipment, can also help lower rising labor costs and boost return on investment (ROI) (SEOAI, 2025). This is due to the high precision wood-working machinery ensuring precision, consistency, and efficiency creating economic and environmental sustainability. This machine relies on “electrical motors to power cutting operations, and their energy efficiency directly impacts operating costs.” According to a study performed by Marconi, the proposed machinery is useful for monitoring defects in products, can be used to augment user knowledge, and allow for increased amounts of energy efficiency throughout the production process (Marconi et al., 2025). In addition, the team found that a 10% reduction in selling price decreases Canada’s NPV relative to the base case highlighting downside risk under adverse pricing condition. This reveals that a Canada-based project would be sensitive to pricing pressure from the market. However, the mitigation investment into CNC automation helps reduce unit labor costs and support margins. By positioning the certified doors as a luxury artisan door, and setting the price point to \$2,450 per door while increasing the price annually, the revenue can be

protected and forecasted to break even by year six. While Canadian timber offers some logistical risk such as regulation, documentation, and seasonality as shown in Table 25, the risks can be mitigated by implementing a high price point, buffer window to account for diversified raw material transportation to account for delays, and utilizing CNC cutting machinery.

Table 26

Risk Matrix: Canada

Canada Risk Matrix				
Risk	Likelihood	Impact	Priority	Mitigation
High domestic labor and administrative cost variables	High	High	High	Invest in CNC automation and maintain premium pricing strategy
Raw material price volatility	Medium	High	High	Secure long-term supplier contracts to diversify sourcing portfolio
Price pressure to due competition in saturated North American market	Medium	High	High	position product as premium artisan door to protect margins and account for high price point
Occasional seasonal supply disruptions	Medium	Medium	Medium	Introduce 30-60 day material transportation buffer when planning for inventory

Note. The above table presents a risk matrix detailing the main risks and associated mitigation strategies for Canada. These risks were formulated using country-specific operating conditions, and the risk priority reflects an assessment based on financial impact. The evaluation of risk and feasibility criteria is as follows: “High” risk indicates a drop in selling price greater than 10% leading to NPV decline or payback period of over 24 months. “Medium” risk indicates that regulatory/documentation delays at the border impacted year-one revenue by 5-8%. “Low” risk indicates that fluctuations in equipment maintenance costs had an impact of less than 2% on annual production.

This was created utilizing information from FSC Connect. (2025). *Monitoring FSC's reach* | FSC Connect. Fsc.org. <https://connect.fsc.org/monitoring-and-evaluation/monitoring-fscs-reach>

Strategic and Ethical Considerations (Tiana)

Strategic and ethical considerations materially influence long-term financial feasibility. While the financial model evaluates projected cash flows under standardized assumptions, sustained value creation depends on supply chain stability, regulatory credibility, and reputational risk management.

Strategic Fit

The proposed export-oriented wooden door manufacturing project aligns with global sourcing strategies that emphasize diversification and supply chain resilience (World Bank, 2021). Recent trade disruptions have demonstrated that lowest-cost production alone does not guarantee long-term stability. Firms increasingly prioritize operational continuity, certification credibility, and regulatory compliance to protect margins and ensure consistent market access (OECD, 2023).

Considering India, Canada, and Nepal provides insight into cost efficiency, institutional stability, and export readiness. Canada offers lower regulatory uncertainty and premium positioning. India provides scalable cost efficiency with moderate risk exposure. Nepal presents lower labor costs but higher logistics volatility. Strategic alignment therefore influences not only market positioning but also the volatility of projected cash flows.

Ethical Considerations

Compliance with international labor and forestry standards is both an ethical obligation and a financial risk management mechanism. Export access to North American and European

markets increasingly requires documented sustainability credentials, including Forest Stewardship Council (FSC) certification and adherence to International Labour Organization (ILO) standards (International Labour Organization, 2021; Cashore et al., 2007).

Sustainable sourcing reduces the probability of regulatory penalties, supply chain interruptions, reputational damage, and contract loss. These risk reductions improve the predictability of future cash flows. In capital budgeting terms, lower volatility and reduced compliance risk may justify a lower risk-adjusted discount rate, thereby increasing Net Present Value (NPV) even if operating margins remain constant

Ethical alignment contributes directly to financial resilience by stabilizing revenue streams, protecting premium pricing opportunities, and reducing long-term capital risk exposure.

Biblical Integration

From a values-based perspective, stewardship principles reinforce sustainable resource management and fair labor practices. Proverbs 11:1 (English Standard Version [ESV], 2016) states, “A false balance is an abomination to the Lord, but a just weight is his delight,” emphasizing integrity in trade. Genesis 2:15 (ESV, 2016) highlights humanity’s responsibility to “work” and “keep” the earth, supporting responsible environmental management. James 5:4 (ESV, 2016) cautions against exploitation of labor.

Applied to export manufacturing, these principles encourage long-term stewardship rather than short-term extraction. Ethical consistency and financial discipline are not contradictory; responsible governance strengthens strategic durability and supports sustainable economic performance.

Conclusion and Recommendations (Lewis)

Summary of Findings

India

The analysis shows strong export market performance in North America and Europe, with India providing a competitive manufacturing base due to cost effectiveness, skilled labor, and developed port infrastructure. Export prices are projected to grow moderately by approximately 2% annually, displaying stable international demand. India exhibits high technical feasibility supported by skilled labor, artisan woodworking techniques, access to semi-automated production technologies and potential to meet international quality standards. Required Infrastructure includes seasoning units, cutting machines, finishing lines, testing equipment, and standardized quality control systems. Higher capacity utilization rates are achievable in India due to relatively mature industrial infrastructure and manufacturing experience. Financial results indicate that an initial Investment of approximately \$5 million is highly viable, generating positive cash flows from the first year, a payback period of about 2 to 3 years. Key financial indicators are highly favorable; Net Present Value indicates \$7,150,688, Internal Rate of Return indicates 24%, and Return on Investment shows 35-40%. Economically, the project supports employment creation, skill development, export growth, and sustainability through responsible sourcing. Although risks aligned with price volatility, raw material costs, and exchange rates exist, sensitivity analysis confirms that the project in India remains financially robust under unfortunate circumstances with effective risk management.

Nepal

The market potential in Nepal exists mainly in mid-range and value-added export sectors, but competitiveness is constrained by higher logistics costs due to the country's landlocked location. Nepal exhibits technical feasibility for wooden door manufacturing based on an

existing woodworking labor force, conventional production methods and a basic industrial base, however automation levels remain limited. Financially, the project requires an initial investment of \$5 million and shows viability with NPV of about \$3.9 million, an IRR of 21%, and a payback period of roughly four years as production becomes sustainable. Economically, the project delivers meaningful benefits through increased job availability, skills upgrading, supplier development, and increased export earnings. Key risks include raw materials supply uncertainty, limited kiln-drying capacity, logistics delays, and sensitivity to export price change. These risks can be mitigated through targeted capital investment, focused product positioning, and improved logistics coordination.

Canada

Canada shows a stable and attractive market base with strong access to North American demand, particularly the United States, benefiting from short delivery times, and the ability to command premium pricing through reliable logistics. The project is technically feasible, supported by advanced industrial infrastructure, automated and semi-automated manufacturing technologies, and a highly skilled workforce that meets strict regulatory and export requirements. Financially, the project requires an initial investment of approximately \$5 million and displays viability with positive NPV of about \$10.3 million, an IRR of 19%, and a payback period of five to six years. Economically, it supports employment, value-added manufacturing, and export earnings while using Canada's developed forestry sector and sustainable production practices. The risks aligned with high labor, raw materials cost, and price sensitivity could be effectively mitigated through automation, premium product positioning, efficient logistics, and diversified sourcing.

Recommendations

Table 27

Comparative Summary Table

Country	NPV (\$)	IRR (%)	Payback period	Major Risk
Canada	10,265,047	19	5 years	High cost of domestic labor and raw materials.
India	7,150,688	24	3 years	Volatility in raw material supply, inconsistent quality, exchange rate fluctuations, and global competition.
Nepal	3,892,864	21	4 years	Timber harvest ban and tight policies.

Based on the technical, market, financial, economic, and risk analyses provided, the project is feasible overall. Across all three country scenarios, the project demonstrates clear technical feasibility, supported by appropriate production technologies and available labor. However, India appears to be the most attractive overall location when comparing the three countries objectively, due to its balance of strong financial performance, technical capability, and manageable risk.

At first glance, Canada appears more attractive because it generates the highest Net Present Value (NPV) of \$10.26 million, compared to India's \$7,150,688. However, NPV reflects absolute value creation rather than return efficiency or capital risk exposure. A more complete

investment appraisal must consider Internal Rate of Return (IRR), payback period, capital recovery speed, and earnings volatility. India provides the highest projected returns, with the highest IRR of 24% and shortest payback period of 3 years, driven by lower labor and operational costs, high-capacity utilization, and efficient access to export ports. Although Canada produces a higher cumulative NPV over time, it locks capital in the project for a significantly longer period and operates under higher fixed labor and raw material cost structures. This increases operating leverage and exposure to cost volatility. In contrast, India's lower production costs and shorter capital recovery cycle reduce financial vulnerability and improve liquidity. From a risk-adjusted investment perspective, India delivers superior capital efficiency and lower downside exposure.

Furthermore, sensitivity analysis confirms that India remains financially robust under adverse scenarios involving exchange rate fluctuations and raw material price changes. Canada, while stable institutionally, carries structurally higher cost risk that compresses margins during downturns. Nepal, despite moderate IRR performance (21%), faces structural logistics and supply chain constraints that elevate operational uncertainty.

Recommendation strategy (Joseph)

Aligning with these findings, the recommended strategy is to locate the large-scale wooden door manufacturing in India while sourcing certified raw wood from Canada. Manufacturing in India allows the company to maximize revenue through cost-efficient production, scalable semi-automated processes, and high utilization rates, which underlines the cash flow improvement and return on investment for the investor. At the same time, getting the FSC-certified timber from Canada leverages the country's well-managed forest resources, strong regulatory oversight, and establishes credibility in North American and European markets. This strategy approach strengthens compliance with sustainability and legal Timber requirements

while supporting premium positioning and buyer confidence. The utilization of India's manufacturing and Canada's raw material sources and sustainability advantages, the company can balance cost-efficiency, risk mitigation, and secure long-term market access.

Conditions and Next Steps

Before finalizing the investment plan, several conditions should be met to ensure project sustainability and reduced risk. Market demand must be confirmed through buyer commitments and competitive pricing analysis. Technical readiness should include access to machinery, skilled labor, and raw material supply, along with necessary certifications. Financial stability must be secured funding, robust cash flow projections, and management of currency or inflation risks. Infrastructure and logistics must guarantee consistent utilities, transport, and storage for production and export. Regulatory compliance, including labor, environmental, and export standards, must be ensured. Furthermore, quality control systems, and a competent management committee should be established to enhance operational efficiency and sustainable returns.

The next steps in the process involve finalizing the feasibility study and securing financing, followed by site selection, regulatory approvals, and land acquisition. Detailed factory design and machineries provision should be completed alongside labor recruitment and training. Pilot production should test processes, quality, and logistics before scaling to full production. At the same time, marketing, distribution, and export operations should be set up, while monitoring and refining processes to maintain efficiency and manage cost. In Conclusion, establishing the wooden door manufacturing project requires careful planning across production, logistics, and marketing, with continuous optimization to ensure long-term profitability. The limitations of this feasibility study include data availability and consistency, Macroeconomic assumptions, Regulatory Simplification, logistics and Infrastructure Variability, and Exclusion of Firm-Level

factors. The leading factors to measure the strengths and weaknesses of each of the countries include Labor cost, manufacturing capacity, Raw materials access, Export infrastructure, Product quality control, and Production cost.

References

- Bhlkaoui, A., & Maksy, M. M. (1986). Basic Needs and Economic Systems: A Sensitivity Analysis. *Review of Social Economy*, 44(2), 178–182. <https://doi.org/10.1080/758527381>
- Brealey, R. A., Myers, S. C., & Allen, F. (2020). *Principles of corporate finance* (13th ed.). McGraw-Hill Education. <https://www.mheducation.com>
- Cashore, B., Auld, G., & Newsom, D. (2007). Forest certification (FSC): *An overview. Annual Review of Environment and Resources*, 32, 117–143.
- Crane, A., Matten, D., Glozer, S., & Spence, L. (2019). *Business ethics: Managing corporate citizenship and sustainability in the age of globalization* (5th ed.). Oxford University Press.
- Denczek, M. (2025, July 7). *Automation Technology for the Window and Door Industry: Transforming Your Production for Peak Efficiency - Stiles Machinery*. Stiles Machinery. <https://stilesmachinery.com/automation-technology-for-the-window-and-door-industry-transforming-your-production-for-peak-efficiency/>
- FSC Connect. (2024). *FSC Risk Assessments | FSC Connect*. Fsc.org. <https://connect.fsc.org/fsc-risk-assessments>
- FSC Connect. (2025). *Monitoring FSC's reach | FSC Connect*. Fsc.org. <https://connect.fsc.org/monitoring-and-evaluation/monitoring-fscs-reach>
- Hassan, M. S., Kausar, A., & Arshed, N. (2022). Investigating Export Determinants: A Time Series Evidence From Canada. *SAGE Open*, 12(2), 215824402211010. <https://doi.org/10.1177/21582440221101037>
- International Labour Organization. (2021). ILO standards on decent work and global supply chains. <https://www.ilo.org/>

- Gibson, L. J. (2012). The hierarchical structure and mechanics of plant materials. *Journal of the Royal Society Interface*, 9(76), 2749–2766. <https://doi.org/10.1098/rsif.2012.0341>
- Himanshu Nitnaware. (2026, January 7). *Union government removes financial and ecological obligations on private plantations in forest areas*. Down to Earth. <https://www.downtoearth.org.in/forests/union-government-removes-financial-and-ecological-obligations-on-private-plantations-in-forest-areas>
- Kanel, K., Shrestha, K., Tuladhar, A., & Regmi, M. (2012). *A Study on The Demand and Supply of Wood Products in Different Regions of Nepal*. Nepal Foresters Association.
- Marconi, S., Frigerio, N., & Matta, A. (2025). Clustering-Based Energy Modeling for Sustainable CNC Woodcutting Machinery. *Procedia CIRP*, 135(32), 307–312. <https://doi.org/10.1016/j.procir.2024.12.024>
- National Association of Home Builders. (2024, August 9). *Home Sizes: Demand vs. Supply*. Nahb.org. <https://www.nahb.org/blog/2024/08/home-sizes>
- National Association of Home Builders. (2026). *Framing Lumber Prices - NAHB*. [Www.nahb.org. https://www.nahb.org/news-and-economics/housing-economics/National-Statistics/Framing-Lumber-Prices](https://www.nahb.org/news-and-economics/housing-economics/National-Statistics/Framing-Lumber-Prices)
- Natural Resources Canada. (2025). Overview of Canada’s forest industry. [https://natural-resources.canada.ca/forest-forestry/forest-industry-trade/overview-c](https://natural-resources.canada.ca/forest-forestry/forest-industry-trade/overview-canadas-forest-industry)
anadas-forest-industry
- OECD. (2020). Global trade and cultural goods. OECD Publishing. <https://www.oecd.org/trade/topics/cultural-goods-and-services/>
- Porter, M. E. (1985). *Competitive advantage: Creating and sustaining superior performance*. Free Press. <https://books.google.com/books?id=H9ReAQAACAAJ>

- SEOAI. (2025, January 27). *How can automated manufacturing reduce errors in door production?* - Pivatic. Pivatic. <https://pivatic.com/en/how-can-automated-manufacturing-reduce-errors-in-door-production/>
- Sharma, R. (2025, November 12). Landlocked to land-linked: Nepal's industrial and logistics transformation. The Annapurna Express. Retrieved February 28, 2026, from <https://theannapurnaexpress.com/story/59934/>
- Statistics Canada. (2023). Sawmill industry statistics and exports. <https://www150.statcan.gc.ca>
- Trivedi, A., Chauhan, S., & Sethy, A. (2018). Bamboo-based composites for sustainable manufacturing applications. *Journal of Cleaner Production*, 204, 318–329. <https://doi.org/10.1016/j.jclepro.2018.08.307>
- United Nations Industrial Development Organization. (2021). Industrial development report 2022: The future of industrialization in a post-pandemic world. UNIDO. <https://www.unido.org>
- United States Census Bureau. (2020). *Business and Industry: Time Series / Trend Charts*. Census.gov. <https://www.census.gov/construction/c30/econ/currentdata/dbsearch?programCode=VIP&startYear=2020&endYear=2026&categories>
- United States Census Bureau. (2026). *Monthly Construction Spending, December 2025*. <https://www.census.gov/construction/c30/pdf/release.pdf>
- World Bank. (2021). Exports, manufacturing, and global value chains. World Bank Publications. <https://www.worldbank.org/en/topic/trade/publication/global-value-chains>

World Bank. (2008, June 16). Landlocked countries: Higher transport costs, delays, less trade [Feature story]. World Bank. Retrieved February 28, 2026, from

<https://www.worldbank.org/en/news/feature/2008/06/16/landlocked-countries-higher-transport-costs-delays-less-trade>

The World Bank. (2023). Nepal country commercial guide. International Trade Administration.

<https://www.trade.gov/country-commercial-guides/nepal-market-overview>